THE GALOIS GROUP OF $x(x-1)\cdots(x-p+1)+1$

CLAUDE (ANTHROPIC)

Abstract. Let p be an odd prime and $f_p(x) = x(x-1)\cdots(x-p+1)+1$. We prove a general certificate: a monic irreducible of prime degree p whose Frobenius at an odd unramified prime q is both odd (Stickelberger) and fixed-point-free has Galois group S_p. The group-theoretic input is that every transitive nonsolvable subgroup of S_p is S_p or lies in A_p (CFSG only through Guralnick). For f_p the fixed-point-free condition is automatic at every $q \le p$, so a single Kronecker symbol $\bigl(\operatorname{disc} f_p/q\bigr) = -1$ with $q < p$ yields $\operatorname{Gal}(f_p/\mathbb{Q}) = S_p$. This verifies the identification for every odd prime $p < 10^5$, and for all primes in six residue classes modulo 36 (Dirichlet density $\frac{1}{2}$). The polynomial is totally real for $p \ge 5$, so conjugation is not a transposition; Newton polygons are flat; φ is Morse in the sense of Serre and $\operatorname{Gal}(\varphi - t/\mathbb{Q}(t)) = S_p$. The remaining obstruction to a proof for every p is whether the periodic good classes of $\bigl(\operatorname{disc} f_p/q\bigr)$, as q varies, cover every prime.

1. The polynomial

Let p be an odd prime, $\varphi(x) = (x)_p = x(x-1)\cdots(x-p+1)$, and

$$f_p(x) = \varphi(x) + 1 \in \mathbb{Z}[x].$$

Reduction modulo p gives $f_p \equiv x^p - x + 1 \pmod{p}$, which is irreducible over $\mathbb{F}_p$ (Artin–Schreier). Hence f_p is irreducible over $\mathbb{Q}$, and $G = \operatorname{Gal}(f_p/\mathbb{Q}) \le S_p$ is transitive of prime degree.

Proposition 1. *If $q \le p$ is prime and $a \in \mathbb{F}_q$, then $f_p(a) = 1$. In particular f_p has no root in $\mathbb{F}_q$.*

Proof. The $p \ge q$ consecutive factors of $\varphi(a)$ cover every residue in $\mathbb{F}_q$, so $\varphi(a) = 0$ and $f_p(a) = 1$. □

Thus Frob_q, for any prime $q \le p$ not dividing $\operatorname{disc} f_p$, is fixed-point-free (Dedekind). At $q = p$ the reduction $f_p \bmod p$ is irreducible, hence separable, so $p \nmid \operatorname{disc} f_p$, and Frob_p is a p-cycle.

2. A general certificate

Theorem 2. *Let $f \in \mathbb{Z}[x]$ be monic irreducible of degree p an odd prime, and let q be an odd prime with $q \nmid \operatorname{disc} f$ such that*

(1) *f has no root in $\mathbb{F}_q$, and*
(2) $\left(\frac{\operatorname{disc} f}{q}\right) = -1$.

Then $\operatorname{Gal}(f/\mathbb{Q}) = S_p$.

The second hypothesis is Stickelberger's parity theorem: $\operatorname{sgn}(\operatorname{Frob}_q) = \bigl(\operatorname{disc} f/q\bigr)$ [2, 1].

Lemma 3. *Let p be an odd prime and $G \leq S_p$ transitive and nonsolvable. Then $G = S_p$ or $G \leq A_p$.*

Equivalently, the only transitive subgroups of S_p containing an odd permutation are S_p and certain subgroups of $\mathrm{AGL}(1, p)$.

Proof of Theorem 2, given Lemma 3. Let $G = \mathrm{Gal}(f/\mathbb{Q}) \leq S_p$, transitive, and $\sigma = \mathrm{Frob}_q$. By Dedekind, σ has no fixed point; by Stickelberger, σ is odd. If G were solvable, Galois's theorem would embed it in $\mathrm{AGL}(1, p) = \{x \mapsto ax + b\}$. An affine map with $a \neq 1$ fixes $b/(1 - a)$, so the only fixed-point-free non-identity elements are translations, i.e. p-cycles, which are even (p odd). This contradicts σ. Thus G is nonsolvable, so Lemma 3 gives $G = S_p$ or $G \leq A_p$; oddness of σ excludes the latter. □

Remark 4 (Sharpness)**.** Neither hypothesis on σ can be dropped. A primitive root in $\mathrm{AGL}(1, p)$ gives type $(1, p - 1)$, sign $(-1)^{p-2} = -1$; translations are fixed-point-free and even. Likewise A_p contains p-cycles. Only the conjunction "odd and fixed-point-free" is impossible outside S_p.

We first isolate the group-theoretic fact needed in the $\mathrm{PSL}_2(11)$ case.

Lemma 5. $\mathrm{PGL}_2(11)$ *has no subgroup of index* 11.

Proof. Suppose $H \leq \mathrm{PGL}_2(11)$ with $|H| = 120$. Since $120 \nmid 660$, one has $H \not\leq \mathrm{PSL}_2(11)$. The image of H in $\mathrm{PGL}_2(11)/\mathrm{PSL}_2(11) \cong C_2$ is therefore nontrivial, so $H/(H \cap \mathrm{PSL}_2(11)) \hookrightarrow C_2$ has order 2 and $H_0 = H \cap \mathrm{PSL}_2(11)$ has order 60. By Dickson's list [5, Hauptsatz II.8.27], the only subgroups of order 60 in $\mathrm{PSL}_2(11)$ are isomorphic to A_5. Then $H \leq N := N_{\mathrm{PGL}_2(11)}(H_0)$ and $N \cap \mathrm{PSL} = N_{\mathrm{PSL}}(A_5) = A_5$ (A_5 maximal and non-normal), so $|N| \in \{60, 120\}$. If $|N| = 120$, the conjugation map $N \to \mathrm{Aut}(A_5)$ has kernel $C_N(A_5)$ meeting A_5 trivially, hence either

- $N = A_5 \times C_2$: then $A_5 \leq C_{\mathrm{PGL}_2(11)}(z)$ for an involution z. Every involution in $\mathrm{PGL}_2(11)$ fixes 0 or 2 points of $\mathbb{P}^1(\mathbb{F}_{11})$, so its centralizer lies in a torus normalizer of order $2(q \mp 1) \in \{20, 24\} < 60$.
- $N \cong S_5$: then N contains F_{20}, in which an element induces an automorphism of order 4 of C_5. But a Sylow C_5 of $\mathrm{PGL}_2(11)$ lies in the split torus and fixes exactly two points, so $N(C_5)$ permutes that pair with kernel the diagonal torus C_{10}: order at most 20, image at most C_2 in $\mathrm{Aut}(C_5) \cong C_4$.

Thus $|N| = 60 < 120 = |H|$, a contradiction. □

Proof of Lemma 3. By Burnside, a transitive group of prime degree is solvable (hence affine) or 2-transitive. So G is 2-transitive, hence primitive. The socle of a 2-transitive group is regular elementary abelian or nonabelian simple; the abelian case is affine, excluded. Thus G is almost simple: $S \leq G \leq \mathrm{Aut}(S)$ with S nonabelian simple, transitive, and $[S : S_\omega] = p$. By Guralnick [4] (cf. Fengler [3, Cor. 2.39]), (S, S_ω) is one of:

(1) $S = A_p$, $S_\omega = A_{p-1}$;
(2) $S = \mathrm{PSL}_d(q_0)$, $p = (q_0^d - 1)/(q_0 - 1)$, S_ω a point or hyperplane stabilizer (d prime);
(3) $S = \mathrm{PSL}_2(11)$, $S_\omega = A_5$, $p = 11$;

(4) $S = M_{11}$ ($p = 11$) or $S = M_{23}$ ($p = 23$).

(The remaining case of Guralnick's theorem, $S = \mathrm{PSU}_4(2)$ with a subgroup of index 27 [3, Thm. 2.38], does not occur: $27 = 3^3$ is a prime power but not a prime.)

The restriction $\mathrm{sgn}|_S$ has kernel of index at most 2; a nonabelian simple group has no subgroup of index 2. So $S \le A_p$ and sgn factors through $G/S \hookrightarrow \mathrm{Out}(S)$.

Case A_p. Then $G \in \{A_p, S_p\}$.

Case M_{11}, M_{23}. $\mathrm{Out}(M_{11}) = \mathrm{Out}(M_{23}) = 1$ [9], so $G = S \le A_p$.

Case $\mathrm{PSL}_2(11)$. Here $\mathrm{Aut}(\mathrm{PSL}_2(11)) = \mathrm{PGL}_2(11)$, so $S < G$ would force $G = \mathrm{PGL}_2(11)$, whose transitive action on the 11 points would require a subgroup of index 11, contradicting Lemma 5. Hence $G = \mathrm{PSL}_2(11) \le A_{11}$.

Case $\mathrm{PSL}_d(q_0)$, $q_0 = r^e$. Let $\Omega = \mathbb{P}^{d-1}(\mathbb{F}_{q_0})$ and take S_ω to be a point stabilizer (composing with the graph automorphism if needed). Distinct points of a 2-transitive group have distinct stabilizers, so if $c \in C_{\mathrm{Sym}(\Omega)}(S)$ then $S_{c(\omega)} = cS_\omega c^{-1} = S_\omega$, hence $c = 1$; the trivial centralizer is used in (c) below. Separately, since $S \trianglelefteq G$, we have $G \le N_{\mathrm{Sym}(\Omega)}(S)$, and every $g \in G$ satisfies $gS_\omega g^{-1} = S_{g\omega}$, so g preserves the S-class of the point stabilizer.

(a) *No diagonal part.* As d is prime, $d \mid q_0 - 1$ would give $p \equiv d \equiv 0 \pmod d$, hence $p = d < 2^d - 1 \le p$, absurd. Thus $\mathrm{PGL}_d(q_0) = \mathrm{PSL}_d(q_0)$.

(b) *No graph part,* $d \ge 3$. The graph automorphism swaps the point- and hyperplane-stabilizer classes [9, §3.3], while field automorphisms preserve the class of a point stabilizer. Class preservation therefore forces $G \le \mathrm{P\Gamma L}_d(q_0)$ and $G/S \le \langle \varphi_r \rangle$.

(c) *The field part is even.* Any $g \in G$ inducing φ_r^k equals $s \cdot \varphi_r^k$ in $\mathrm{Sym}(\Omega)$ with $s \in S \le A_p$, so $\mathrm{sgn}(g) = \mathrm{sgn}(\varphi_r)^k$. If e is odd, φ_r has odd order, all cycles have odd length, and φ_r is even. This covers all $d \ge 3$: if e were even, $s_0 = r^{e/2}$ would give

$$p = \frac{s_0^d - 1}{s_0 - 1} \cdot \frac{s_0^d + 1}{s_0 + 1}$$

with both factors integers > 1 (d odd). For $d = 2$, $p = q_0 + 1$ prime forces $q_0 = 2^e$ with $e = 2^j$ (Fermat). On $\mathbb{P}^1(\mathbb{F}_{2^e})$, φ_2 fixes $\{0, 1, \infty\}$ and every nontrivial orbit has 2-power length ≥ 2, hence is a cycle of even length and an odd permutation, so $\mathrm{sgn}(\varphi_2) = (-1)^t$ with t the number of nontrivial orbits. The number of orbits of length 2^i is the number of monic irreducibles of degree 2^i over $\mathbb{F}_2$,

$$n_i = \frac{2^{2^i} - 2^{2^{i-1}}}{2^i}, \qquad v_2(n_i) = 2^{i-1} - i.$$

Thus $n_1 = 1$ and $n_2 = 3$ are odd, while n_i is even for $i \ge 3$. For $j \ge 2$, $t \equiv n_1 + n_2 \equiv 0 \pmod 2$ and φ_2 is even. For $j = 1$, $\mathrm{P\Gamma L}_2(4) \cong S_5 = S_p$.

In all cases $G = S_p$ or $G \le A_p$. $\square$

CFSG enters only through Guralnick's theorem. The facts $\mathrm{Out}(M_{11}) = \mathrm{Out}(M_{23}) = 1$ and non-conjugacy of the two parabolic classes for $d \ge 3$ are in [9]; maximality of A_5 in $\mathrm{PSL}_2(11)$ is in Dickson's list [5, Hauptsatz II.8.27]. The involution-centralizer bound and the absence of F_{20} in $\mathrm{PGL}_2(11)$, and the signs of φ_2 on $\mathbb{P}^1(\mathbb{F}_{2^e})$ for $e = 2, 4, 8, 16$, are finite checks. (If an order-4 element conjugated an element of order 5 to its cube, its inverse would conjugate it to its square, so testing $bab^{-1} = a^2$ over all order-4 elements covers both generators of $\mathrm{Aut}(C_5)$.)

3. The certificate for f_p

Corollary 6. *Let p be an odd prime. If some odd prime $q < p$ satisfies $\left(\frac{\operatorname{disc} f_p}{q}\right) = -1$, then $\mathrm{Gal}(f_p/\mathbb{Q}) = S_p$.*

Proof. Hypothesis (1) of Theorem 2 holds for every prime $q \le p$ by Proposition 1, so Theorem 2 applies. □

The prime $q = p$ can never serve: Frob_p is an even p-cycle, equivalently $\operatorname{disc} f_p \equiv (-1)^{(p+1)/2} \pmod p$ is always a residue. Evaluating the symbol at a candidate $q < p$ is a single resultant over $\mathbb{F}_q$. For such q,

$$f_p \equiv (x^q - x)^{\lfloor p/q \rfloor} \prod_{k < p \bmod q} (x - k) + 1 \pmod q,$$

and $\operatorname{disc} f_p \bmod q$ is the discriminant of this reduction; the symbol is its quadratic character. A search over odd primes $q < p$ finds a witness $q \le 47$ for every prime $7 \le p < 10^5$ (largest witness $q = 47$ at $p = 31511$; $q = 3$ succeeds for 50.2% of 9589 primes, matching the density $\frac{1}{2}$ of the classes in Proposition 10). For $p = 5$ the only candidate is $q = 3$, and $\big(38569/3\big) = +1$, so the certificate has no witness; S_5 is settled instead by the type-$(2,3)$ Frobenius at $q = 19$. For $p = 3$, $\operatorname{disc} f_3 = -23$ is not a square, so $G = S_3$.[1]

4. Closed routes

Proposition 7 (Totally real)**.** *For every prime $p \ge 5$, f_p has p real roots. (For $p = 3$ there is one real root.)*

Proof. Write $x_k = k - \frac{1}{2}$ for $0 \le k \le p$. The sign of $\varphi(x_k)$ is $(-1)^{p-k}$, and

$$\frac{|\varphi(x_{k+1})|}{|\varphi(x_k)|} = \frac{k + \frac{1}{2}}{p - k - \frac{1}{2}}.$$

The sequence $|\varphi(x_k)|$ is therefore strictly unimodal with minimum at the centre $m = (p-1)/2$,

$$\min_k |\varphi(x_k)| = \frac{(2m-1)!!\,(2m+1)!!}{2^p}.$$

This equals $45/32 > 1$ at $p = 5$ and thereafter grows by the factor $(2m+1)(2m+3)/4$. Hence $f_p = \varphi + 1$ has the sign of φ at all $p + 1$ half-integers, so p sign changes and p real roots. At $p = 3$ the minimum is $3/8 < 1$. □

Complex conjugation is therefore not a transposition for $p \ge 5$, and Conrad's criterion (irreducible of prime degree, exactly two non-real roots) does not apply.

[1] The complete witness list for $7 \le p < 10^5$, the $p \le 97$ discriminant data, the $q = 3$ and $q = 5$ class tables of §5, and the verification programs are included with this note (`ancillary/` for data, `tools/` for the checkers, one per computational claim). Each row (p, q) of the witness list is verifiable independently of the search that produced it, by one resultant and one Legendre symbol over $\mathbb{F}_q$.

Proposition 8. *The polynomial φ is a Morse function in the sense of Serre* [8, §4.4]*: the zeros of φ' are simple and the finite critical values are pairwise distinct. Consequently* $\mathrm{Gal}(\varphi - t/\mathbb{Q}(t)) = S_p$ [8, Thm. 4.4.5].

Proof. On each interval $(k, k+1)$, $k = 0, \ldots, p-2$, the logarithmic derivative $H(x) = \sum_{j=0}^{p-1}(x-j)^{-1}$ is strictly decreasing ($H' = -\sum(x-j)^{-2} < 0$), so there is a unique simple zero c_k. Thus $\varphi''(c_k) = \varphi(c_k)H'(c_k) \neq 0$. (The point ∞ is a totally ramified critical point of multiplicity p, excluded from Serre's definition.) The involution $x \mapsto (p-1) - x$ sends c_k to c_{p-2-k} and φ to $-\varphi$.

Write $\mu_k = \max_{(k,k+1)} |\varphi|$ and $m = (p-1)/2$. For $x \in (k, k+1)$ one has $|\varphi(x+1)/\varphi(x)| = (x+1)/(p-1-x)$, which is < 1 throughout $(k, k+1)$ whenever $k \leq m-2$. Thus $|\varphi(x+1)| < |\varphi(x)|$ pointwise on that interval, and taking maxima yields $\mu_{k+1} < \mu_k$ for $k = 0, \ldots, m-2$. The central pair has $\mu_{m-1} = \mu_m$ by the involution, so equal magnitudes occur only for mirror pairs $c_k \leftrightarrow c_{p-2-k}$. Those indices have opposite parity (p odd), hence opposite signs $(-1)^{p-1-k}$; since no critical value vanishes (the roots of φ are simple), the two members of a mirror pair are distinct. Critical values of different magnitude are distinct a fortiori, so all finite critical values are pairwise distinct. □

(The specialisation $t = -1$ can still be exceptional: $(x)_4 + 1 = (x^2 - 3x + 1)^2$. Hilbert irreducibility of the family does not decide this single value.)

Newton polygons of f_p are flat: $f_p(0) = \cdots = f_p(p-1) = 1$. The same holds for every translate $f_p(x+a)$ with $0 \leq a < p$, which is what excludes the method. Hilbert–Grunwald and effective HIT prescribe or bound *some* specialisation of $\varphi - t$, never the fixed $t = -1$.

The arithmetic Morse criterion of [8, Remark 2, pp. 42–43] (at most one double root modulo every prime; inertia generates Gal by Minkowski; the worked example $x^n - x - 1$, following Nart–Vila [6] and Osada [7]) applies to f_p itself once $\operatorname{disc} f_p$ is squarefree. Direct factorisation gives this for $p \leq 7$; already at $p = 97$ the discriminant has 12642 digits. Among $p \leq 97$, every ramified prime $\ell \leq 10^6$ has $v_\ell(\operatorname{disc} f_p) = 1$. A single such ℓ produces an inertia transposition and, with the mod-p p-cycle, yields S_p; this fires for 18 of the 24 primes $p \leq 97$.

5. Periodicity of the discriminant

The same identity that makes Frob_q fixed-point-free computes the Stickelberger symbol at a fixed odd prime $q < p$. Write $p = mq + r$ with $0 < r < q$, and set $g = x^q - x$, $B_r = (x)_r$. Then $\varphi \equiv g^m B_r \pmod{q}$. Differentiating and using $g' \equiv -1$ gives

$$\varphi' \equiv g^{m-1}\psi \pmod{q}, \qquad \psi = gB_r' - mB_r.$$

The polynomial ψ has degree at most $q + r - 1 \leq 2q - 2$ and depends on p only through r and $m \bmod q$. Multiplicativity of the resultant together with Proposition 1 yields $\operatorname{Res}(g, f_p) = \prod_{a \in \mathbb{F}_q} f_p(a) = 1$, so the factor g^{m-1} contributes 1^{m-1} and

$$\operatorname{disc} f_p \equiv (-1)^{(p-1)/2} \operatorname{Res}(\psi,\, g^m B_r + 1) \pmod{q}. \tag{1}$$

The unbounded part of the discriminant has disappeared. What remains is periodic.

Lemma 9 (Periodicity). *Let q be an odd prime and*

$$\Pi(q) = \operatorname{lcm}\Big(4,\ q^2,\ \operatorname{lcm}_{1\le d\le 2q-2}(q^d-1)\Big).$$

For odd primes $p, p' > q$, if $p \equiv p' \pmod{\Pi(q)}$ then $\operatorname{disc} f_p \equiv \operatorname{disc} f_{p'} \pmod q$, and in particular $\big(\operatorname{disc} f_p/q\big) = \big(\operatorname{disc} f_{p'}/q\big)$.

Proof. The right-hand side of (1) is defined for any odd integer $n = mq+r > q$ not divisible by q, via $f_n = (x)_n + 1$. Writing $\operatorname{lc}(\psi)$ for the leading coefficient of ψ,

$$\operatorname{Res}(\psi, f_n) = \operatorname{lc}(\psi)^n \prod_{\psi(\beta)=0} f_n(\beta)^{m_\beta},$$

the product over roots in an algebraic closure, with multiplicity. The sign $(-1)^{(n-1)/2}$ has period 4. The polynomial ψ itself is determined by r and $m \bmod q$, hence by $n \bmod q^2$; this is the only source of the factor q^2 in $\Pi(q)$. The power $\operatorname{lc}(\psi)^n$ depends on $n \bmod (q-1)$, and $q-1$ divides the $d=1$ term of the inner lcm.

Each factor $f_n(\beta) = g(\beta)^m B_r(\beta) + 1$ is an evaluation in a field. If $\beta \in \mathbb{F}_q$ then $g(\beta) = 0$, so $f_n(\beta) = 1$. If $\beta \notin \mathbb{F}_q$ then $g(\beta) \neq 0$, because every root of g lies in $\mathbb{F}_q$, and $g(\beta)^m$ has period $\operatorname{ord}\big(g(\beta)\big)$ dividing $q^d - 1$ where $d = [\mathbb{F}_q(\beta) : \mathbb{F}_q] \le 2q-2$.

If $n \equiv n' \pmod{\Pi(q)}$, these data agree. (For the multiplicative orders: $\gcd(q, q^d-1) = 1$, so $n \equiv n'$ modulo q^2 and modulo $\operatorname{lcm}_d(q^d-1)$ implies $m \equiv m'$ modulo that lcm.) Restricting to primes yields the lemma. $\square$

The bound is not always sharp: $\Pi(3) = 9360$ while the actual period is 36, and $\Pi(5)$ is enormous while the period is $600 = \operatorname{lcm}(4, 5^2, 5^2-1)$. The inner lcm cannot be dropped in general, because ψ can have complementary factors of degree greater than 1. For $q=3$ the six polynomials ψ can be read off by hand, and among odd p the residue $\operatorname{Res}(\psi, f_p)$ is constant on each class modulo 9.

Proposition 10. *Let $p > 3$ be prime. Then $\big(\operatorname{disc} f_p/3\big) = -1$ if and only if*

$$p \equiv 7, 13, 17, 19, 23, 29 \pmod{36}.$$

Proof. Write $p = 3m + r$ with $r \in \{1,2\}$ and $g = x^3 - x$. The six values of ψ as $r \in \{1,2\}$ and $m \bmod 3$ are as follows. In each case $\operatorname{Res}(\psi, f_p)$ is a constant $R \in \{1,2\}$ on odd p, recorded with the corresponding class $p \bmod 9$.

$r=1$. Here $B_1 = x$, so $\psi = g - mx = x\big(x^2 - (m+1)\big)$.

- $m \equiv 0 \pmod 3$: $\psi = g$. All roots lie in $\mathbb{F}_3$, so $f_p(\beta) = 1$ at every root and $R = 1$. Then $p \equiv 1 \pmod 9$.
- $m \equiv 2 \pmod 3$: $\psi = x^3$, so $R = f_p(0)^3 = 1$. Then $p \equiv 7 \pmod 9$.
- $m \equiv 1 \pmod 3$: $\psi = x(x^2+1)$. Let $\alpha^2 = -1$ in $\mathbb{F}_9$. Then $g(\alpha) = \alpha(\alpha^2 - 1) = \alpha$, so

  $$R = \big(\alpha^{m+1} + 1\big)\big((-\alpha)^{m+1} + 1\big).$$

  Odd $p = 3m+1$ forces m even, hence $m+1$ odd, and the product is $1 - (\alpha^2)^{m+1} = 1 - (-1) = 2$. Then $p \equiv 4 \pmod 9$.

$r=2$. Here $B_2 = x(x-1)$ and $B_2' = 2x-1 = 2(x+1)$ in $\mathbb{F}_3$, so

$$\psi = x(x-1)\bigl(2(x+1)^2 - m\bigr).$$

- $m \equiv 0 \pmod 3$: $\psi = 2x(x-1)(x+1)^2$, all roots in $\mathbb{F}_3$, so $f_p(\beta)=1$ at every root and $R = 2^p \equiv 2$. Then $p \equiv 2 \pmod 9$.
- $m \equiv 2 \pmod 3$: $\psi = 2x^2(x-1)^2$, likewise $f_p(\beta)=1$ at every root and $R = 2^p \equiv 2$. Then $p \equiv 8 \pmod 9$.
- $m \equiv 1 \pmod 3$: $\psi = 2x(x-1)(x^2+2x+2)$. The quadratic is irreducible (disc $\equiv 2$, not a square). For a root β one has $\beta^2 = \beta+1$, $g(\beta) = \beta(\beta^2-1) = \beta^2$, and $B_2(\beta) = \beta^2-\beta = 1$, so $R = 2^p N_{\mathbb{F}_9/\mathbb{F}_3}\bigl(1+(\beta^2)^m\bigr)$. Now $(\beta^2)^2 = (\beta+1)^2 = 2$, hence β^2 has order 4 in $\mathbb{F}_9^\times$. Odd $p = 3m+2$ forces m odd, so $(\beta^2)^m \in \{\beta^2, -\beta^2\}$; both $1+\beta^2 = \beta-1$ and $1-\beta^2 = -\beta$ have norm 2. Thus $R = 2^p\cdot 2 \equiv 1$. Then $p \equiv 5 \pmod 9$.

The Kronecker symbol is $\bigl((-1)^{(p-1)/2}R/3\bigr)$. Since $\bigl(1/3\bigr)=1$ and $\bigl(2/3\bigr) = -1$, this is $(-1)^{(p-1)/2}$ when $R=1$ and the opposite when $R=2$. Combining the class modulo 9 with the sign modulo 4 yields the six listed classes modulo 36, and no others among the units of $\mathbb{Z}/36\mathbb{Z}$. $\square$

The good set is stable under $p \mapsto -p$ (this fails already for $q=5$). Dirichlet's theorem supplies infinitely many primes in each of the six classes, and Corollary 6 with $q=3$ yields $\mathrm{Gal}(f_p/\mathbb{Q}) = S_p$ on a set of Dirichlet density $\frac{1}{2}$. In particular:

Theorem 11. *There are infinitely many primes p with* $\mathrm{Gal}(f_p/\mathbb{Q}) = S_p$.

This theorem is computer-free: it uses only Theorem 2 with Lemma 3 (the two finite checks in that proof are hand-verifiable), Proposition 10, and Dirichlet's theorem.

Each further prime q contributes its own periodic good set. For $q=5$, an inspection of the twenty polynomials ψ shows that the part coprime to g has degree 0, 2, or 4, so Lemma 9 applies with the inner lcm restricted to $d \in \{1,2,4\}$ and yields period dividing $15600 = \mathrm{lcm}(4, 5^2, 5^4-1)$. Evaluating (1) at one representative of each odd class modulo 15600 shows that the period divides 600, and no proper even divisor of 600 works. Among the 160 units of $\mathbb{Z}/600\mathbb{Z}$, 88 have symbol -1. Combining with Proposition 10: of the 480 units of $\mathbb{Z}/1800\mathbb{Z}$, $372 = 240+264-132$ are good for $q=3$ or $q=5$ — the six classes modulo 36 lift to 240 units modulo 1800, the 88 classes modulo 600 lift to 264, and the intersection has 132 units. As $132 = 480\cdot\frac{1}{2}\cdot\frac{11}{20}$, the two good sets are exactly independent on the units modulo 1800, which is not automatic since $\gcd(36,600)=12$.

Theorem 12. *The set of primes p with $\bigl(\operatorname{disc} f_p/3\bigr) = -1$ has Dirichlet density $\frac{1}{2}$. The set of primes with $\bigl(\operatorname{disc} f_p/3\bigr) = -1$ or $\bigl(\operatorname{disc} f_p/5\bigr) = -1$ has Dirichlet density $\frac{31}{40}$. On both sets,* $\mathrm{Gal}(f_p/\mathbb{Q}) = S_p$.

The second density uses the $q=5$ representative evaluation; the first does not. Both match the least-witness counts below 10^5: $q=3$ succeeds for 50.2% of 9589 primes, and $\min(q)\le 5$ for 77.6%.

6. What remains

Write $\operatorname{disc} f_p = (-1)^{(p-1)/2} \operatorname{Res}(\varphi', \varphi + 1)$. Reducing modulo p (where $\varphi' \equiv -1$) gives the congruence

$$\operatorname{disc} f_p \equiv (-1)^{(p+1)/2} \pmod{p},$$

always a quadratic residue, as forced by the even p-cycle Frob_p. This is a consistency check, not a constraint on witnesses $q < p$.

The identification $\mathrm{Gal}(f_p/\mathbb{Q}) = S_p$ for every odd prime p is now a covering question. A finite list of q would finish all large p, but the data already refutes covering by $q \leq 43$: the prime $p = 31511 \equiv 11 \pmod{36}$ has least witness 47. The least witness appears unbounded, so all-p is unlikely to fall to a fixed finite set of congruences. What remains is to show that no prime survives in the (explicit, shrinking) bad classes at every level $q < p$.

Two features make this sharper than asking for the square class of an unstructured discriminant. The good sets are explicit congruence classes, computed once per q. (The $q = 3$ symmetry $p \mapsto -p$ already fails for $q = 5$, so there is no functional equation of that form across q.) Any analytic argument that $\operatorname{disc} f_p$ is a nonresidue at some $q < p$ need only treat primes in the complementary classes $1, 5, 11, 25, 31, 35 \pmod{36}$, a density-$\frac{1}{2}$ family, rather than all p. (The cases $p = 3, 5$ are settled by other witnesses, as above.) The covering, and the identification for every p, have been verified for all odd primes $p < 10^5$.

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